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NB-IoT optimization for smart meters networks of smart cities: Case study

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Abstract The future demands for Smart Meters (SM) applications impose maximizing resources utilization. In this paper, we achieve two main objectives to comply with fifth-generation networks goals and serving the SM applications. First one is to enhance the Narrowband-Internet-of-Things spectral efficiency. The second is to alleviate the signalling burden during each transmission request for SMs. First, we model the uplink scheduler for the NB-IoT access network using the state-machine modelling methodology on the Simulink environment. The simulation results are verified and validated with trusted modelling technique. Second, we optimize the NB-IoT uplink scheduler, which exploits the periodicity nature of SMs applications. We achieve that by proposing an integrated scheduling protocol which rearranges the transmission times of different smart meters utilities, and it draws a map for the transmission schedule of them to better utilize the sparse time resources. The proposed approach comprises architecture, signalling, and algorithms. We evaluate the optimization in term of numbers of SMs versus four performance indicators; resources utilization, spectral efficiency, setup success percentage, and session drop rate according to a case study. The simulation results prove the ability of our proposal to increase the spectrum utilization to 17.47%. This enhancement reflects on spectral efficiency that improved doubled.

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1. Introduction

1.1. IoT 5G/5G + Networks

Internet of Things (IoT) applications are one of the emerged technology applications that are considered a key digital transformation enabler for several industries and utilities, including smart cities, factories, electricity grids, many other services.

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Recently, the IoT would be used as a component in the operational technologies of nuclear power plants and radiation monitoring networks. Therefore, they need a reliable and efficient network infrastructure for optimum operation.

Contrary to human traffic, the IoT communication profile is completely different from what is experienced by smartphones over fourth-generation (4G) cellular networks. Where, a sufficient number of intelligent objects will become ubiquitous, enabling them to sense and manipulate information in real-time. Meanwhile, smartphones sessions start according to the human profile. For periodic applications, a device transmits the sensed data periodically to a network to deliver innovative and real-time services [1]. For example, at smart cities, the smart meter does not only register periodic reading but also it draws the profile for the object resources consumption. Where, this profile can be used for future predictions of seasonal energy needs. So, smart meters play an important role in decision making at different levels. Consequently, the SMs became one of the most used and prevalence applications in IoT domain [2–4].

The increased population density worldwide and the evolution of human and community lifestyles have led to the rapid development of mobile communication technology. Thus, the 3rd Generation Partnership Project (3GPP) has standardized the vision for 5th generation mobile communication technology (5G), which comprises the traditional LTE network alongside the future 5G network. The 5G and the 5G+ mobile networks target to overcome the previous cellular standards limitations and to become an enabling technology for future IoT applications. The 3GPP releases of 5G, starting from release 15 and beyond, introduced enhanced cellular communications services such as enhanced mobile broadband (eMBB) and massive machine-type communications (mMTC), and they also introduced new services such as ultra-reliable low latency communication (URLLC) and time-sensitive communications (TSC) [5–7]. The URLLC and mMTC in 5G are closely related to IoT [7]. Compared to the current 4G, it is expected that 5G mobile networks will efficiently support the basic IoT requirements, such as good coverage, high data throughput, low latency, high scalability, high energy efficiency, and the ubiquitous connectivity provision for IoT end-users [5]. 5G aims to maximize the capacity of the network to be 1000 times larger than 4G [8].

The IoT 5G umbrella includes current technologies such as unlicensed bands (short-range communication technologies including Bluetooth, Zigbee, Wi-Fi, and the unused spectrum band such as cognitive radio [9] and low-power wide-area networks (LPWANs). The latter are divided into non-3GPP networks such as SigFox, LoRa, [3,4], and 3GPP networks such as Long Term Evolution (LTE)-M in Release 12 and narrowband-IoT (NB-IoT) in Release 13 [4,10–17]. Recently, Fog radio access network (Fog-RAN) and Cloud radio access network (C-RAN) have been proposed for fifth-generation (5G) wireless communications that serve IoT applications [18]. For smart city applications, LPWANs are the most suitable alternatives [3]. Therefore, we only focus on 5G cellular IoT in this paper.

NB-IoT achieves unrivalled LPWA performance in terms of coverage and the variety of data rates and deployment scenarios compared to other LPWA technologies [19]. This technology is a modified design of LTE that aims to serve IoT traffic. Besides, it able to provide enhanced coverage compared to LTE. It is used in three modes: standalone, in-band, and guard-band [20]. The subcarrier bandwidth is 180 kHz in the case of co-existence with current LTE networks to provide long periods of connections over mobile operator networks because various objects have small amounts of data [20]. The aim of NB-IoT is globally covering low-cost or complex ubiquitous IoT devices worldwide [21].

To cater for the massive IoT requirements, the 3GPP has standardized that the mMTC services will be served by further evolving NB-IoT and LTE-M as integral parts of the 5G specifications, such that there will be no need for new standards in the near future [22]. Although they are known as 4G technologies, the 3GPP has started a working group in the 5G core network to support LTE-M and NB-IoT radio access networks in order to guarantee the compatibility of these technologies with 5G [22].

1.2. Problem statement

Efficient and profitable IoT communication systems should be low cost, size, weight, and power enabled. With the increased dependence on NB-IoT technology in people's lives, a large number of NB-IoT terminals need to be connected to the network in order to cater to the various needs of users. According to predictions by Ericsson, the number of connected IoT devices will reach 1.5 billion by 2022 [23].

The rapid increase in the number of massive IoT devices has added a growing challenge to 5G design owing to its contrasting requirements [8] from two main perspectives. First, the narrow bandwidth may restrict the transmission performance of the IoT in addition to the communication overhead that is required before data transmission [8]. Therefore, it is important to achieve continuous improvements concerning the efficient use of these resources to support the continued growth of IoT applications [24]. Second, access reservation protocols designed for 5G networks face massive connectivity for IoT applications [3], because the device density is expected to be larger than the capability of the techniques to realize access reservation and RA procedures, and may significantly affect end-to-end delay. This issue would be critical, especially for the scenarios of burst arrival [25]. The access requests and the signalling associated with each periodic IoT transmission process are a burden to both IoT device and network resources. These problems are further transferred with uplink traffic and remain opened issues, as discussed in [20,26].

Until now, 5G systems have been considered an evolution of the LTE-advanced (LTE-A) standard, where the 5G new radio (NR) access reservation is also based on LTE-A random access (RA) procedure [25,27]. Access class barring (ACB) and its modifications is a standardized technique to process burst access in LTE and NR RA [25]. However, it may not be appropriate for IoT applications [1]. Most adopted solutions focused on random-access phase optimization and other optimization techniques, as we will present in Section 2. Currently, studies and developments for the NB-IoT have been widely adopted by global mobile operators and manufacturing companies [24].

1.3. Related works

The optimization works in MAC layer improve spectral and energy efficiency to serve IoT traffic. This paper will adopt

spectral efficiency approach. The researches of this approach adopted many different directions as follow.

The authors of references [28] adopted the random accessing stage in MAC to improve the system throughput by using Markov chain modelling technique. In [29], Begishev et al. studied three BW allocation strategies; Static Allocation, Dynamic Allocation, and Dynamic Allocation with reservation between LTE and NB-IoT. The study recommended the "dynamic resource sharing with reservation" strategy for heterogeneous IoT traffic. The evaluation process was executed through formulating a model by using a two dimensional Markov chain. The first improvement trial for resources utilization was introduced in [30]. Where, Bing-Zhi et al. designed an uplink (UL) scheduling algorithm that sort NB-IoT UEs based on minimum scheduling delay. Their algorithm was built by using a Software Defined Network (SDN). The persistent scheduling strategy for periodic IoT applications was adopted for the first time in [31]. Although it was addressed from an energy efficiency perspective, the authors focused only on proposing prediction algorithm foresees the periodicity of IoT traffic without reviewing the messaging procedures and their impact. Chen et al. [32] optimized the NB-IoT uplink scheduling using the improved K-mean algorithm, and they considered the IoT distribution model as an additional impact on random accessing and its consequences on system throughput. On the contrary, in [33], the periodic IoT traffic was addressed from a different perspective that focuses only on alleviate the signalling burden. This perspective achieved by eliminating some RRC procedures and only in limited cases. The NB-IoT group communication technique was adopted clearly in [34]. Their proposal was based on adapting Multimedia Broadcast Multicast Service (MBMS) protocol for serving downlink traffic of IoTs. The cashing based group access was adopted in [8] for Fog-RAN and in [35] for Industrial IoT covered by small base stations. Jin in [36] introduced group based accessing of same location and attributes under the group leader. B. M. ElHalawany et al. in [37] maximized the cell throughput via adopting the Device to Device (D2D) communication methodology. However, their clustering idea was based on uniform deployment condition.

All the works, as mentioned above, did not consider the periodicity nature of periodic IoT applications. Therefore, the issue of improving the system efficiency is still opened due to the traffic management researches is still weak and insufficient the study for the nature of IoT traffic types. This paper presents an integrated vision to optimize this issue.

1.4. Paper contribution

This paper has two main objectives. The first one is to enhance the NB-IoT spectral efficiency through increasing its spectrum utilization. The second objective is to reduce the signalling burden during each transmission request for periodic IoT applications. We achieved these objectives through the following two steps:

- First, we model NB-IoT uplink scheduler. The model is executed using the state-machine modelling methodology over Simulink. We employ Simulink's built-in features, which provide continuous run at different simulating durations as desired for mathematical and state-machine models. These features achieve by adopting different differential and integral equations types and techniques such as [38–40], which were added to recent Simulink releases.
- In the second step, we present an approach that employs the persistent scheduling strategy based on the periodicity nature of periodic IoT applications to develop integrated scheduling protocol. The proposal handles three modification axes signalling, architecture, and processing algorithm. The proposal modified the signalling by adding IoT type identification during Radio Resources Control (RRC) connection establishment messaging phase. At architecture axis, the medium access control (MAC) layer was chosen to execute the approached algorithm. The third axis represents processing algorithm.

Our algorithm is quite simple to alleviate the processing burden for both the SM terminal and the access node. For getting fair results, we divide the cell spectrum of NB-IoT equally on four data rates. We apply our proposal on one of the most famous periodic IoT applications that are smart meters. The optimization is applied to a case study for a designated geographical location has smart meters for two different utilities. The optimization is evaluated in term of the number of NB-IoT UEs (in our case study, we call SM terminals or SM UEs) versus four KPIs; resources utilization, spectral efficiency, RRC setup success percentage, and session drop rate. The simulation results prove the ability of our proposal to increase the spectrum utilization to 17.47%. This enhancement has a positive impact on spectral efficiency, which improved doubled.

1.5. Paper organization

The remainder of the paper is organized as follow. Section 2 will describe our model design, our proposed optimization and smart meters case study with its traffic profiles. The analysis and the discussion for simulation results will be presented in Section 3. Finally, our conclusion will be presented in Section 4.

2. NB-IoT uplink scheduler model

This section describes our modelling methodology, our optimization approach and applying the optimization on smart meters case study at three parts as follow. The first one describes the model design of the NB-IoT uplink scheduler. The second subsection describes our optimization approach for NB-IoT uplink scheduler. The case study is described in the third subsection.

2.1. Model description

In general, the MAC layer charges of four main functions: random access control, uplink and downlink UEs data scheduling control, Hybrid Automatic Repeat reQuest (HARQ) process, and finally mapping between logical and transport channels [41]. Fig. 1 explains the message sequence chart (MSC), starting from random access request up to uplink data scheduling grant according to 3GPP release 14 [42,43]. In the figure, the messages sequencing are from one to seven.

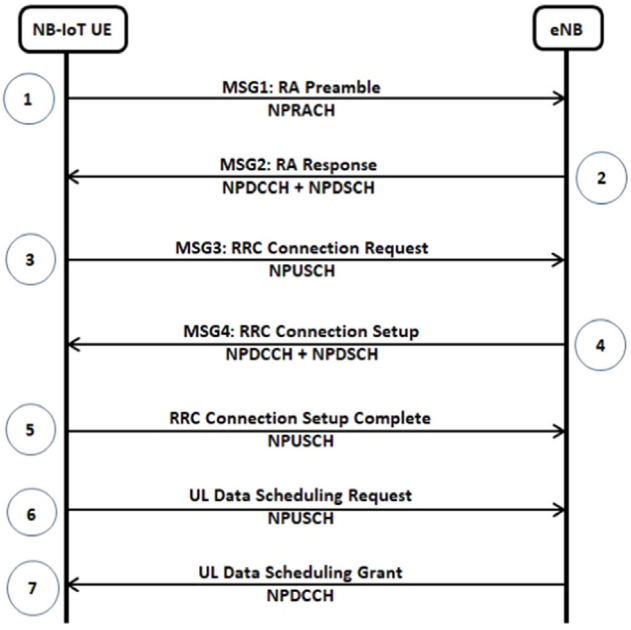


Fig. 1 MSC between NB-IoT UE and eNB from RRC connection establishment phase to data transfer.

Accordingly, this MSC is divided into three main super-states. The MAC layer sub-block in NB-IoT UE entity consists of three main super-states; random access phase, RRC messaging phase, and uplink data scheduling phase. The MAC layer state-machine model of eNB consists of four super states as depicted in Fig. 2. Where, there are two super-states for uplink data scheduling, one for NB-IoT band tones and another super-state for LTE band resources (it is considered for future works). The four super-states are parallel executions to provide concurrent uplink services to all traffic model types according to the traffic profile of each traffic model. Each super-state consists of sub-states that model the process sequences of the function of its super-state. Where, in the "RRC messaging phase" super-state of Fig. 2, it consists of five sub-states as depicted in Fig. 3. Each sub-state models a message sequence of Fig. 1 with the same order. Also in Fig. 2, the super states numbered three and four respectively model the messages number six and seven of Fig. 1. According to the addressed topic in this work, we focus on modelling the uplink scheduling and its associated functions.

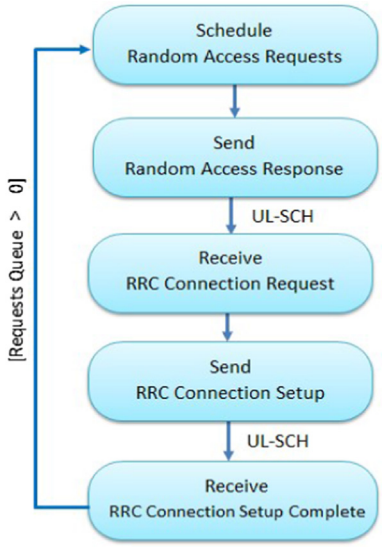


Fig. 3 eNB State level of "RRC Messaging Phase" sub-state.

Also, the smart meters applications traffic model with its traffic profile and density was modelled in state-machine methodology as presented in Fig. 4 using the state flow toolbox of Simulink. The SMs model block consists of two super-states. The first one performs session initiation, traffic density, and traffic profile. The second super-state manages the payload size transfer after receiving acceptance from the RRC layer.

2.2. Proposed optimization

We exploit the periodicity of periodic applications, where, the SMs applications have a uniform profile. Therefore, our enhancement proposal benefits from this feature to enhance the utilization of the NB-IoT spectrum. We realized this idea by proposing an integrated scheduling protocol that adopts the persistent scheduling strategy to rearrange the transmission times of periodic data of different utilities and to draw a map for their transmission schedule. Accordingly, we adequately utilize the free radio frame resources. The optimization approach comprises a combination of three axes: architecture, signalling, and algorithms. Each axis represents a modification and addition to standardized NB-IoT by 3GPP [42,43].

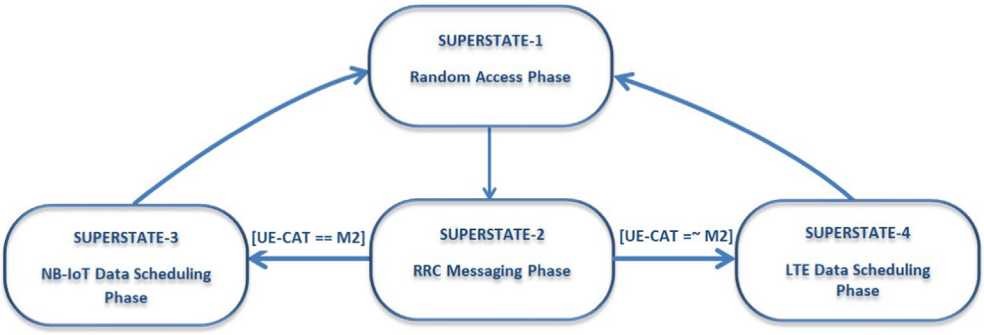


Fig. 2 NB-IoT eNB MAC layer state machine model.

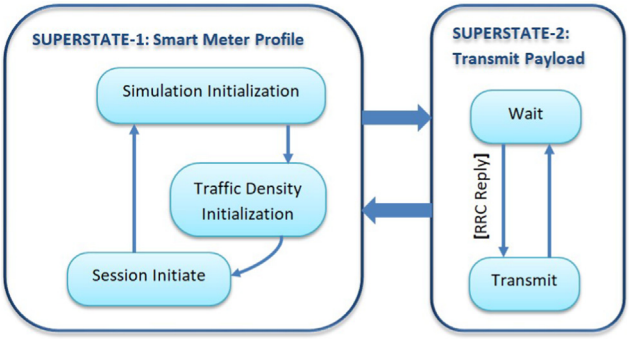


Fig. 4 Smart Meter sub-block Simulink model – State level.

The proposed approach classifies SM devices in such a geographical coverage area like as our case study into groups or clusters, as shown in Fig. 5. This classification accords to the type of utility or service provided by these smart meters. For example, groups of smart meters transmit periodic meter readings that are related to water, gas, power consumption, and any other types of periodic metering services that have their periodicity transmission. We use the word "cluster" because this group of SM terminals is associated with a specific utility; consequently, all of these terminals will initiate random access requests simultaneously. In our case study, we apply our proposal on two different utilities; electricity utility and water utility.

2.2.1. Architecture

The architectural change focuses mainly on the MAC layer (layer 2) of the protocol as it represents the executive manager for radio resource management in addition to the feasibility and flexibility to apply a modification at this layer compared with the physical layer. The designed algorithms work on an

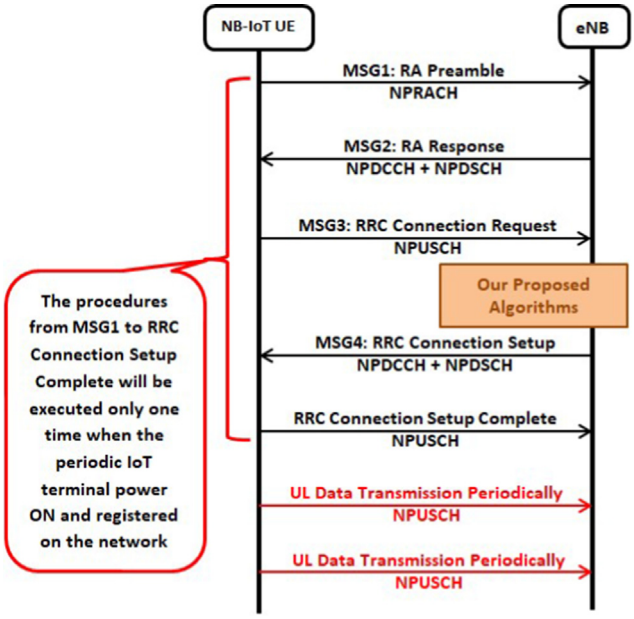


Fig. 6 MSC between NB-IoT UE and eNB after optimization.

RRC CONNECTION REQUEST message and RRC CONNECTION SETUP message, as depicted in the MSC of Fig. 6. After the eNB receives the cluster information that is carried in the RRC CONNECTION REQUEST message, it replies to the cluster using the RRC CONNECTION SETUP message.

2.2.2. Signalling

The ultimate goal of this research is to ensure optimum data capacity communications that is suitable for one of the most

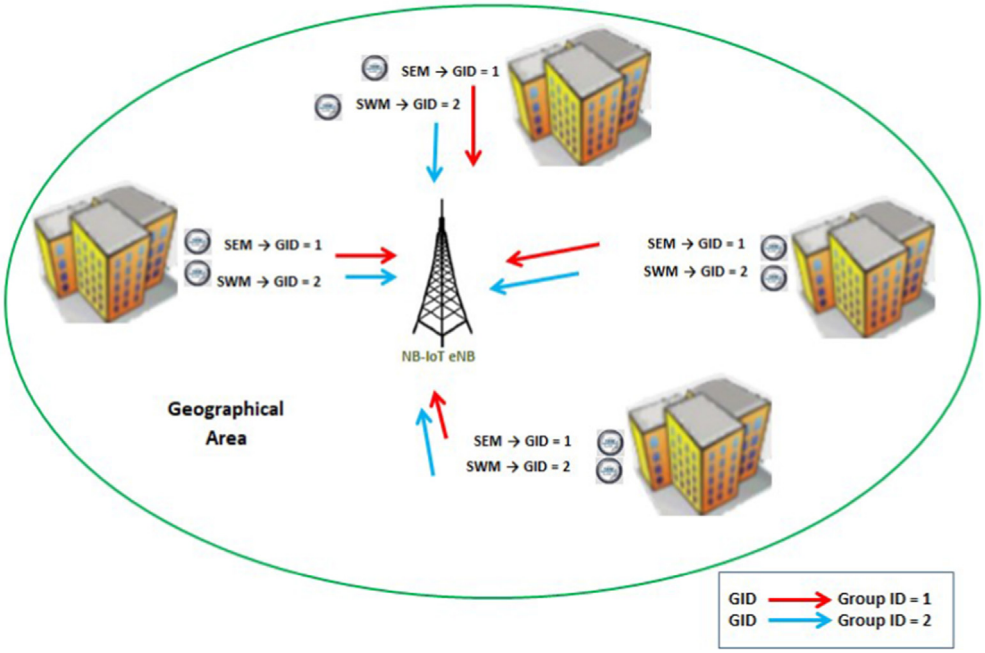


Fig. 5 Geographical area has SMs, and each SM of utility has GID.

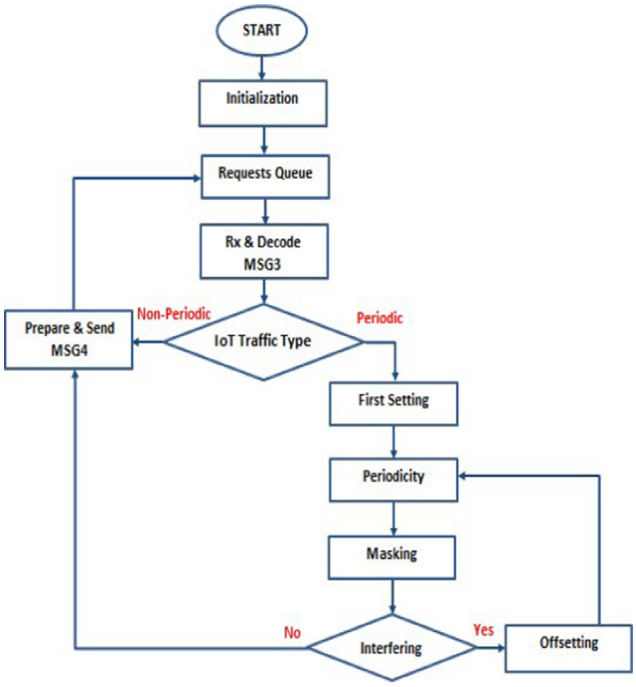


Fig. 7 The overall proposed algorithms flow chart.

used and widespread applications at IoT industry like smart metering, and which is economically feasible as well as spectrally efficient. Accordingly, it is necessary to add more functions to the signalling dictionary, which could enable negotiation between nodes to achieve efficient operations in addition to the architectural changes proposed.

The proposed changes summarize in identifying the following four new information fields that will be carried by the RRC CONNECTION REQUEST:

(1) ***IoT-Type:*** determines the profile of the NB-IoT UE, i.e., whether or not it is periodic,
(2) ***Cluster-ID:*** a new ID identifies the utility of the cluster,
(3) ***Cluster-Periodicity:*** identifies the information about the periodicity of the cluster with a millisecond time unit,
(4) ***Cluster-Payload:*** determines the payload size per NB-IoT UE.

Subsequently, the execution of the proposed algorithms draws the cluster transmission map. Then, the synchronization with the network will reply to the NB-IoT UE via the RRC CONNECTION SETUP msg in one additional information field and will be called cluster-synchronization. The proposed messages to be modified are related to 3GPP [42].

2.2.3. Algorithm

In our proposed procedures, the first task is initialization. Where, we create a Clusters Time Table, which formulated as a matrix as depicted in Eq. (1). This matrix represents a database for all SM clusters that will be registered on the access network. Consequently, these SM terminals will avoid the random access and the RRC connection establishment procedures during the next periodic data transmissions. Then, the first condition to run the rest of our proposed procedures is to determine the IoT traffic type. As previously mentioned, our proposed approach focuses on periodic IoT traffic. As depicted in Fig. 7, the overall proposed procedures consist of four phases, each of which depends on the previous one and has its algorithm. We model our proposed procedures in the RRC messaging phase by adding two sub-states have a red colour between the MSG3 and MSG4 sub-states, as depicted in Fig. 8 during the transporting of the RRC messages via the MAC layer as well as after decoding and encoding the required information fields. These algorithms are implemented using MATLAB scripts that are linked as files with our Simulink project. Fig. 9 presents the realization of the proposed model

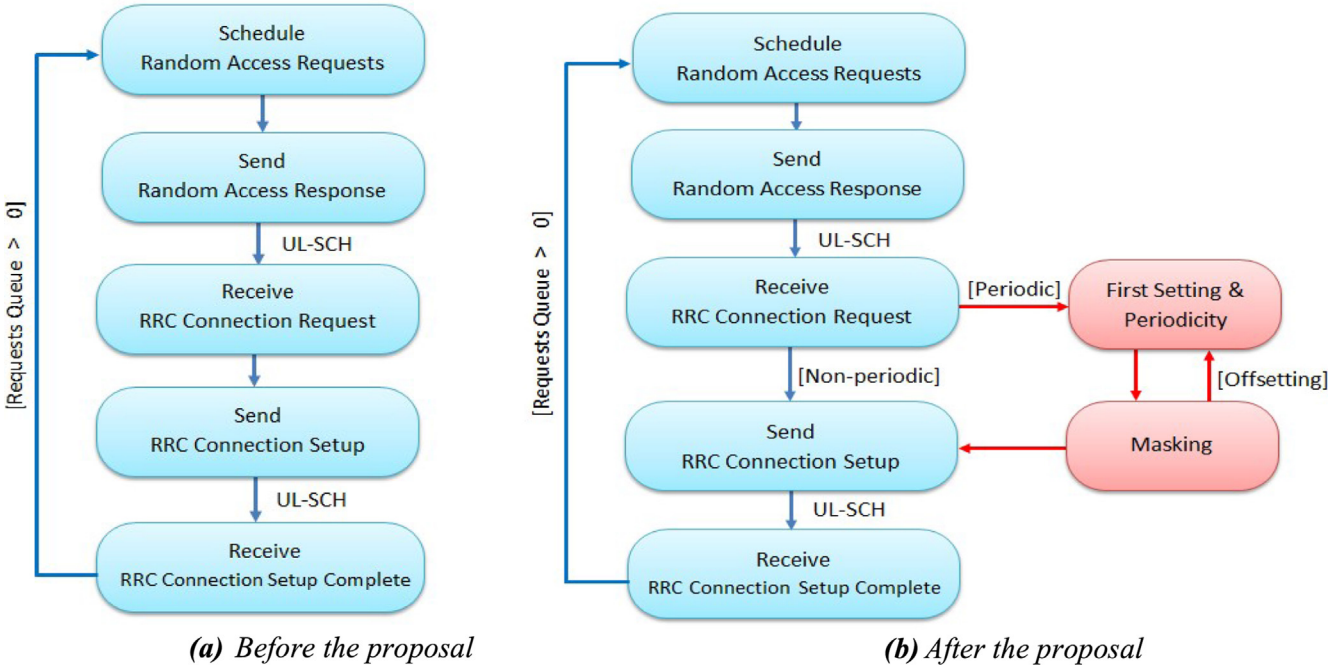


Fig. 8 The comparison of the State level of RRC Messaging Phase before and after our proposal.

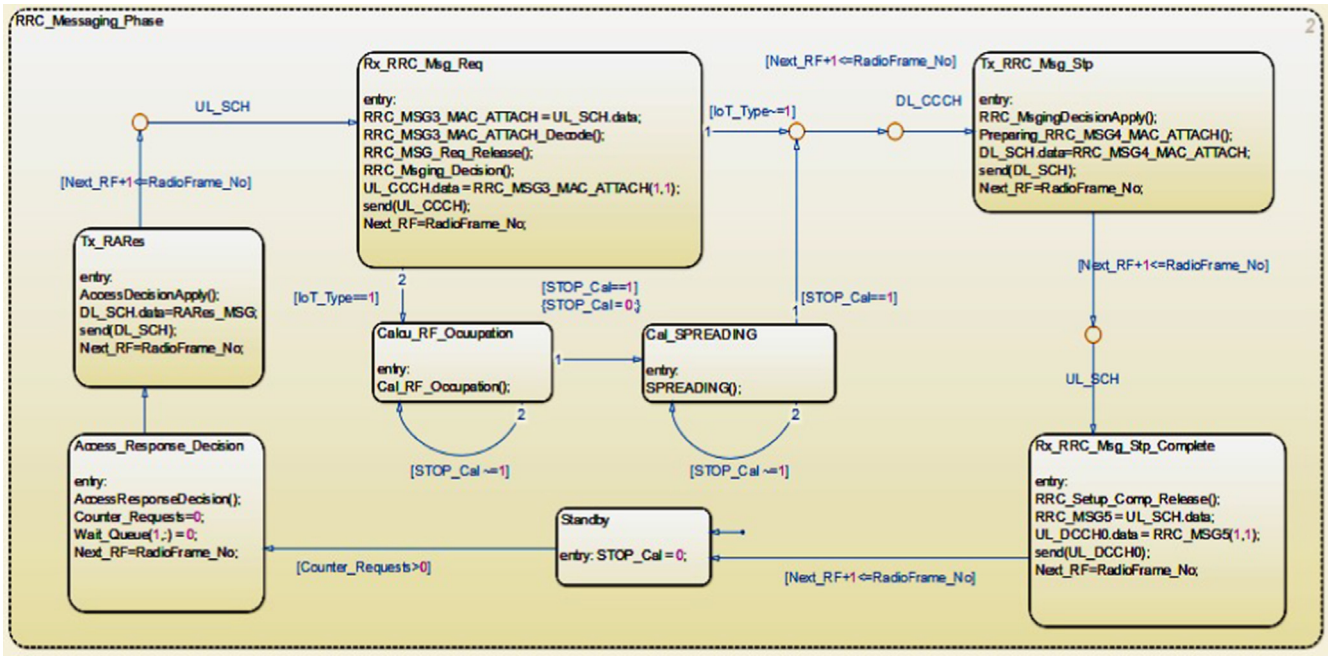


Fig. 9 Modelling of "RRC Messaging Phase – after proposal" using State flow toolbox of Simulink.

using State flow toolbox of Simulink. Table 1 describes the notations in each algorithm. In the following section, we present a description of the algorithm of each phase.

- **Initializations:** the base station sets and creates the following variables:
- Clusters Time Table Index = 1
- Create Transmission Map array
- Create Clusters Time Table matrix [4 × n]

$$Clusters\ \ Time\ \ Table = \begin{bmatrix} C_{ID} & \cdots & C_n \\ C_p & \cdots & C_n \\ U_{rf} & \cdots & C_n \\ t_{Cn} & \cdots & C_n \end{bmatrix} \quad (1)$$

Table 1 List of Notations.

Notation	Value
C_{ID}	Cluster-ID that is received by MSG3
C_P	Cluster periodicity that is received by MSG3
U_{rf}	Occupied radio frame
t_{Cn}	Synchronization time of the cluster No. n with eNB
p	Payload size of IoT terminal that is received by MSG3
Cn_{start}	Start time of cluster data transmission
Cn_{stop}	Stop time of cluster data transmission
N_{CP}	Periodicity counts during one hour
N_{RF}	Radio frame count of cluster periodicity

- **Phase 1: First Setting** – during this phase, the start and end of the first periodic transmission is determined and to accomplish the synchronization with the network. The transmission duration and transmission end time are determined according to the cluster payload.

Algorithm 1: First Setting

Input: p,
Output: Cn_{start} , Cn_{stop}
1: Compute U_{rf} , U_{rf} = Buffer-Size-Indexing (p)
2: if *Clusters Time Table Index* = 1 **then**
3: $t_{C1} = t_o$, where $n = 1$
4: else $t_{Cn} = t_{Cn-1} + U_{rf-1}$
5: end if
6: $Cn_{start} = t_{Cn}$
7: $Cn_{stop} = t_{Cn} + U_{rf}$

- **Phase 2: Periodicity** – according to the calculation that was done during the first phase (cluster transmission duration and transmission end), the next periodic transmissions are determined over one hour (1-h) based on the cluster periodicity that was given in RRC message 3. The cluster periodicity is the intergeneration time (IGT) for SM terminals. In this phase, the IGT is translated to the number of radio frames. Accordingly, the "Cluster Map" over a 1-h period is created.

Algorithm 2: Periodicity

Input: C_P , C_{ID} , Cn_{start} , Cn_{stop}
Output: *cluster Map* array [1, $Cn_{start} \rightarrow Cn_{stop}$]
1: N_{CP} = One Hour duration / C_P
2: while $n_{CP} > N_{CP}$ **do**
3: Create *cluster Map* array [1, $Cn_{start} \rightarrow Cn_{stop}$] = C_{ID}
4: *cluster Map* (1, Cn_{start}) = $Cn_{start} + N_{RF}$
5: *cluster Map* (1, Cn_{stop}) = $Cn_{stop} + N_{RF}$
6: $n_{CP} = n_{CP} + 1$
7: end while

- **Phase 3: Masking** – according to the cluster map that was created in the previous phase, a masking or comparison process between the cluster map and NB-IoT eNB transmission map execute to detect any time interference with any other registered cluster transmission schedule during the hour. If the masking results indicate free interfering, then the algorithm returns to the waiting state to await any new cluster transmission scheduling requests.

Algorithm 3: Masking

Input: U_{rf}
Output: *Offset-Value*
1: Initialize *Offset-Value* = 0
2: [*MASK*] = find the U_{rf} numbers in *Cluster MAP*
3: for $i = 1 \rightarrow$ length(*MASK*) **do**
4: **if** *Transmission MAP* (1, *MASK* (1, *i*)) < 0 **do**
5: *Offset-Value* = *Offset-Value* + 1
6: **else**
7: **end if**
8: end for

- **Phase 4: Offsetting** – this phase begins depending on the results of the masking process in the previous phase. If any time interference with any other registered cluster is detected, the offsetting process is done for the first periodic transmission. Then, the algorithm returns to the periodicity phase for verification. This process repeats itself until it is free from interference.

Algorithm 4: Offsetting

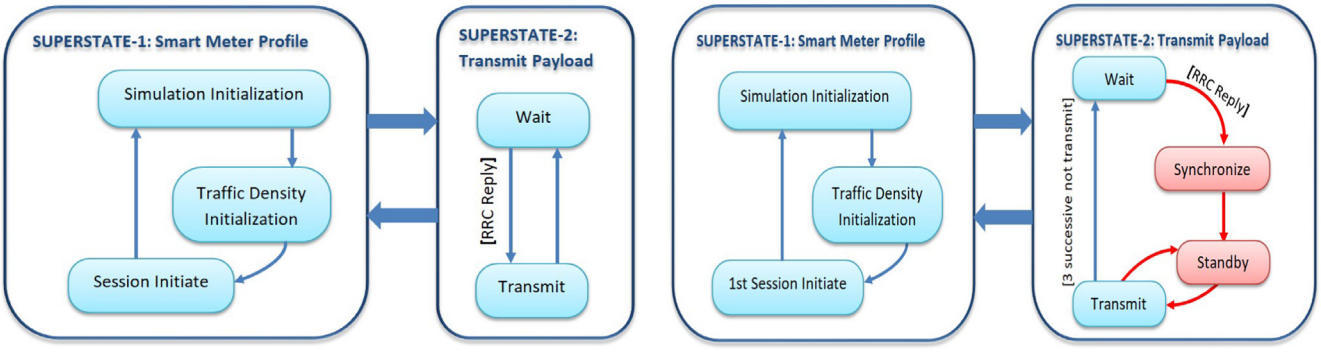
Intput: *Offset-Value*
1: if *Offset-Value* > 0 **then**
2: $Cn_{start} = Cn_{start} + Offset\text{-}Value$
3: $Cn_{stop} = Cn_{stop} + Offset\text{-}Value$
4: **go to Algorithm 2**
5: else update *Transmission MAP* with *cluster Map n*
6: end if
7: update *Clusters Time Table*
8: *Clusters Time Table Index* = *Clusters Time Table Index* + 1
9: go to Prepare MSG4

Also, the smart meter traffic model was slightly modified as depicted in Fig. 10(b). Where, the session initiation procedures are executed once upon first registration in the network. Then, the next periodic transmission times will be managed by the second super-state after coordination with eNB.

2.3. Smart meters case study

We apply our case study on the urban location of district number six in Rehab County in New Cairo city – Egypt, which is shown in Fig. 11(a) and (b) from satellite picture google earth [44]. The area of this location is approximately 200 km^2. So, this location can be covered by one NB-IoT eNB site according to Refs. [45,46]. This district comprises 145 residential building. Where, each one has 20 apartments. Each apartment has two types of smart meters for two different utilities. The first one is a Smart Electricity Meter (ESM), and the second one is a Smart Water Meter (SWM). Consequently, the total numbers of the Smart Meters (SM) are 5800 in this district.

The traffic profiles of ESM and SWM are according to Refs. [47,48] respectively. However, we modified the intergeneration time (IGT) of SWM from 60 Minute to 15 Minute. Table 2 describes the traffic profile for these SMs.



(a) Before the proposal *(b) After the proposal*

Fig. 10 The comparison of the Smart Meter sub-block Simulink model – State level before and after our proposal.



(a) (b)

Fig. 11 The geographical location of the case study, (a) Satellite picture and (b) Google Map Layout depicts the exact total location area and its perimeter distance.

Table 2 Simulation Parameters.

Parameter	Value
eNB Bandwidth	20 MHz
Total Number of RBs	100
Number of RBs allocated to NB-IoT band	80
Number of RBs allocated to Single Tone	20
Number of Single Tone Resource Units	240
SEM payload	381 Byte
SWM payload	115 Byte
SWM packet count	5
Intergeneration Time "IGT"	15 min

3. Simulation results analysis and discussion

In this section, we describe our simulation setting and model conditions and verify our model. Then we analyze and discuss our simulation results for our optimization.

3.1. Simulation setup

First, the simulation state flow of RRC connection establishment and data scheduling procedures is verified according to 3GPP standard release 13 [39,40] and as depicted in MSC of Fig. 1. Moreover, our model results are validated to the standard results of NB-IoT scheduling process which, it is built by the ns-3 simulator in [32]. With the difference that our simulation runs duration is 60 min. The round-robin scheduling technique is adopted. We model two smart meters groups, "SWM" Group of the water utility and "SEM" Group of the electricity utility access simultaneously the NB-IoT base station. We simulate three scenarios, where the first and second scenarios, (A & B), simulate the SWM and SEM respectively and based on the random access, RRC connection establishment, and data scheduling procedures of the 3GPP standard [42,43]. The third scenario, (Scenario C), simulates our proposed approach. The simulation parameters are described and summarized in Table 2. For fairness results, the cell spectrum of NB-IoT is divided equally on the four data rates. A Uniform traffic density is assumed with linear increasing at each simulation run and propagation without transmission impairments. We use the state flow toolbox of Simulink (Rel.2016).

3.2. Proposed optimization results analysis

This paper improves NB-IoT spectral efficiency. According to the achieved improvement, our model is measured from four aspects as follow; Utilization, Throughput, Accessibility, and Retainability. Its equations statistics are explained in [49] specifically for radio network KPIs and confirmed with 3GPP standard in [50,51]. We applied our optimization scheme using the frequency band that is allocated for NB-IoT single-tones, as depicted in Table 2. In order to verify the efficiency of our proposed optimization, we compared our results with the 3GPP standard method [42,43] graphically and with the improved K-mean method (IKMM) [32] numerically in the discussion section.

3.2.1. Utilization

Fig. 12 presents a snapshot of 1 s duration versus the utilization of the NB-IoT single-tone resource spectrum. Fig. 12

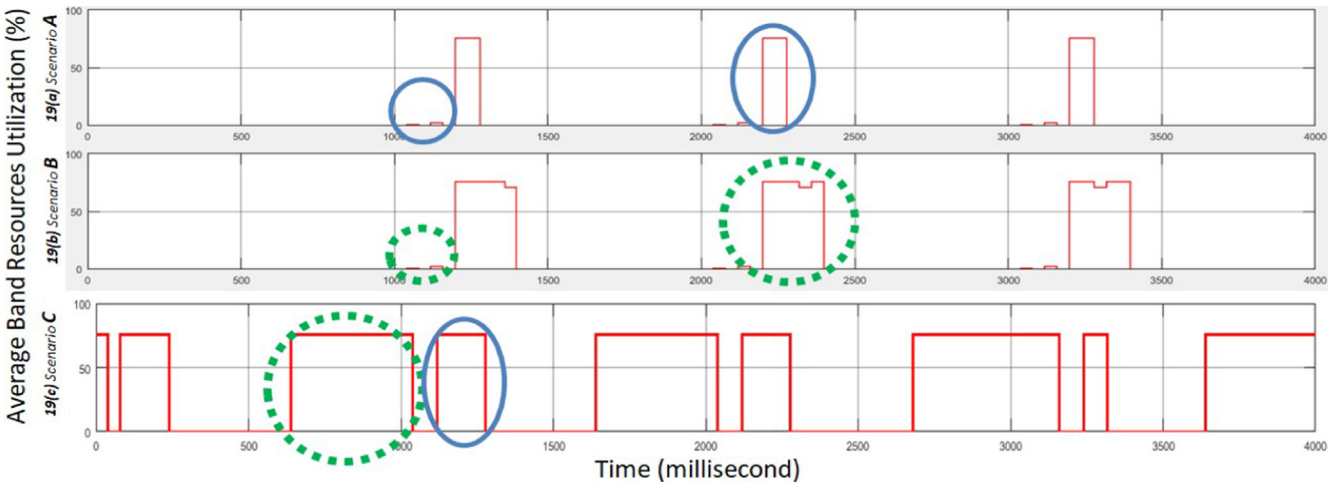


Fig. 12 The screenshot depicts the comparison between the three scenarios for band resources reservation of data channels during one second, (A) Traffic of SEM group by current scheduling scheme, (B) Traffic of SWM group by current scheduling scheme, and (C) Traffic of SEM group & SWM group in proposed scheduling scheme.

(a)–(c) shows the three simulated scenarios. As previously mentioned, the first and second scenarios follow the standard, and the third follows our proposed scheme. According to our model, the first scenario shows that all SWM Group UEs only succeeded to dominate all single-tone bands, and all SEM Group UEs failed. The small blue solid circle represents the allocated resources for uplink RRC messaging. Then, the large blue solid circle represents the allocated resources for the uplink payload. The second scenario shows that all SEM Group UEs only succeeded to dominate all single-tone bands, and all SWM Group UEs failed. The small green dotted circle represents the allocated resources for uplink RRC messaging, and the large green dotted circle represents the allocated resources for the uplink payload. In Fig. 12(a) and (b), both the small and large red squares represent the tones and its radio frames that are used to carry the control messages and data payload. From Fig. 12(a) and (b), we observe free radio frames, which represent unutilized tones during the rest of the second duration. The third scenario presents the allocated channels for both groups and the allocated radio frames for each group after applying our proposal. Fig. 12(c) shows that our proposed approach utilized the resources wasted for signalling and higher layer control messaging at each transmission process. These resources were used instead in the transfer of more periodic IoT traffic. Also, the proposed approach maximized the utilization of radio frame resources.

Fig. 13 presents the enhancement in the NB-IoT band single-tone utilization with respect to the number of periodic NB-IoT UEs (in our case study are called SM terminals or SM UEs) during a 1-h (one hour) duration, where for the SWM Group traffic that uses the standard method the spectrum saturated at 9.88%. Moreover, even when the payloads in the SEM Group traffic increased with using the standard, the spectrum saturated at 22.05%. After applying our proposed approach, the spectrum saturated at 39.52%. Consequently, the increase in channel utilization reached 17.47%.

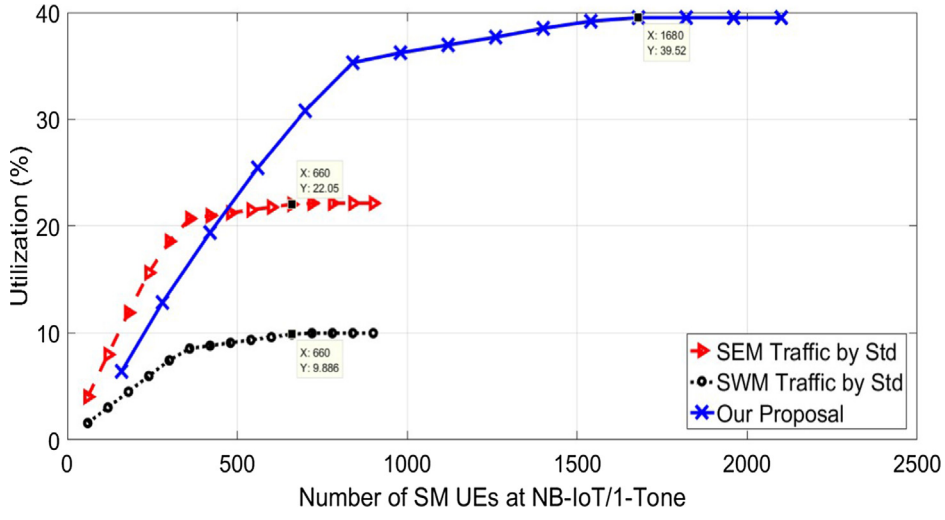


Fig. 13 Comparison of NB-IoT Resources Utilization for the current standard and the proposed scheme.

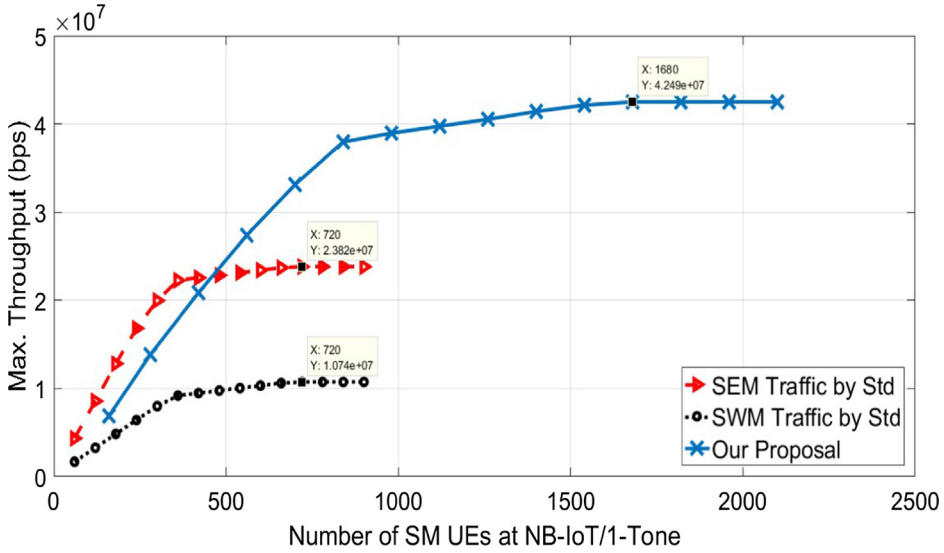


Fig. 14 Comparison of NB-IoT system throughput for current standard and the proposed scheme.

3.2.2. Throughput

The utilization enhancement reflects on NB-IoT system throughput. Fig. 14 depicts the system throughput versus increasing the numbers of SM UEs. For SWM Group payloads, the throughput is 10.7 Mbps. Even with an increased payload in SEM Group, the maximum achievable throughput is 23.82 Mbps when using the standard. After applying our proposed scheme, the maximum achievable throughput becomes 42.49 Mbps. Fig. 15 presents the relation between the spectral efficiency and the increased numbers of SM UEs, where the maximum spectral efficiency is 5.95bb/s/Hz for the SEM Group traffic when using the 3GPP standard. The maximum spectral efficiency becomes 10.62b/s/Hz with our proposed approach. Additionally, the number of SM UEs that can be served until the NB-IoT spectrum reach to saturation is depicted in Fig. 16. Where, the numbers of SM terminals doubled after our proposal, which they increased from 720 terminals to 1680 after the proposal.

3.2.3. Accessibility

Fig. 17 presents the RRC Setup Success percentage versus increasing the numbers of SM UEs. The figure depicts the enhancement of the RRC Setup Success percentage from 660 SM terminals to 100% success at 1540 SM terminals.

3.2.4. Retainability

Fig. 18 represents the relation between Session Drop Rate (SDR) and increasing the numbers of SM UEs. Also, improving the utilization contributed to save and increase the resources not only for more smart meters devices but also to complete their session. Where, at our proposed optimization, the SDR starts

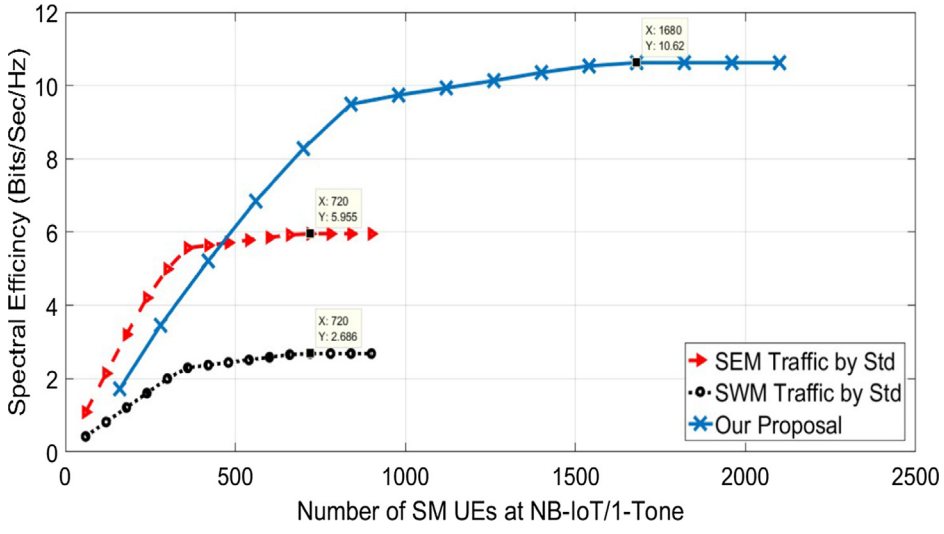


Fig. 15 Comparison of NB-IoT Spectral efficiency for current standard and the proposed scheme.

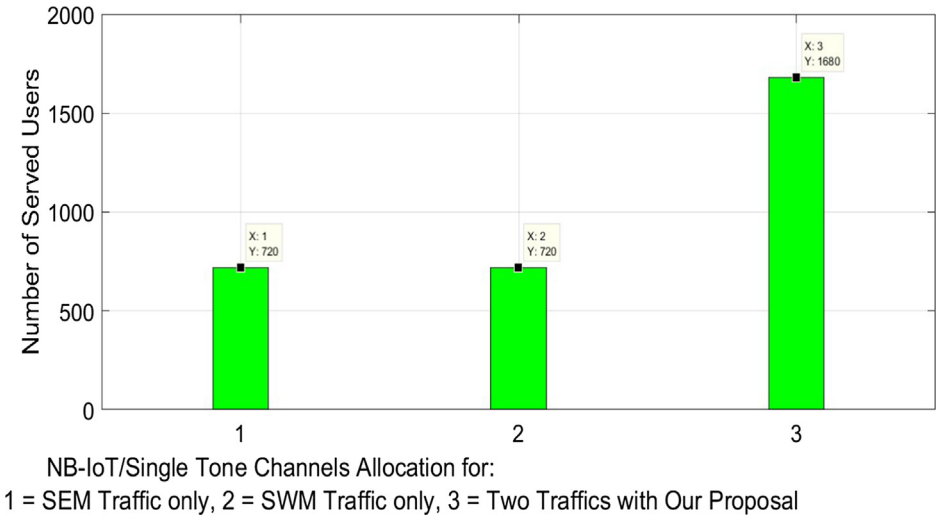


Fig. 16 Comparison of maximum numbers of served NB-IoT UEs for current standard and the proposed scheme.

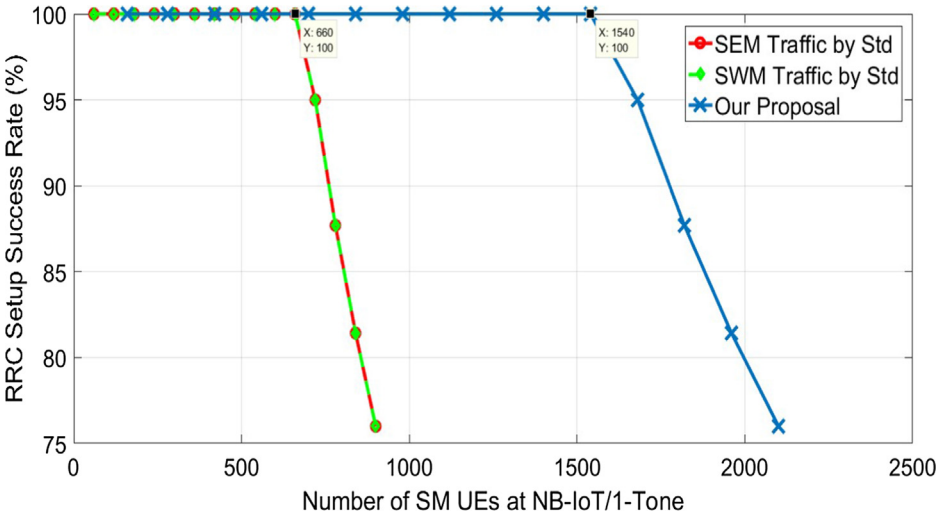


Fig. 17 Comparison between RRC Setup Success Rate for the current standard and the proposed scheme.

at 560 devices with the lowest SDR rate, as depicted in the figure, instead of 240 devices at standard procedures.

3.3. Results discussion

In this paper, we exploited the persistent scheduling strategy and the periodicity nature of the periodic IoT application like smart meter. We realized that by proposing an integrated scheduling protocol which rearranges the transmission times of different smart meters utilities and draws a map for the transmission schedule of them. As a result, the comprehensive utilization of free time resources achieves. Our proposal succeeded to improve three main KPIs based on real scenarios as follow.

(1) The first main KPI is the NB-IoT system bandwidth utilization percentage during one second, which is enhanced by 17.47% according to the traffic condition of the case that was studied. Where, the highest utilization achieved was 22.05% then after the proposal became 39.52%. The enhancement in the utilization strongly influenced on system throughput and spectral efficiency.

a. From the system throughput aspect, it is improved from 23.82 Mbps to 42.49 Mbps at 1680 SM terminals after our proposal. Comparing to the optimization in [32] which the highest achievable throughput was below 500bps at 5000 IoT terminals with 20Byte for the packet size whereas at our traffic condition the smallest packet size was 23Byte [43]. Our proposal operated on 4 MHz bandwidth comprises 228 NB-IoT single tones of 15 kHz bandwidth.
b. From the spectral efficiency aspect, it is improved doubled. Where, it was 5.95b/s/Hz then became 10.62b/s/Hz after our proposal.

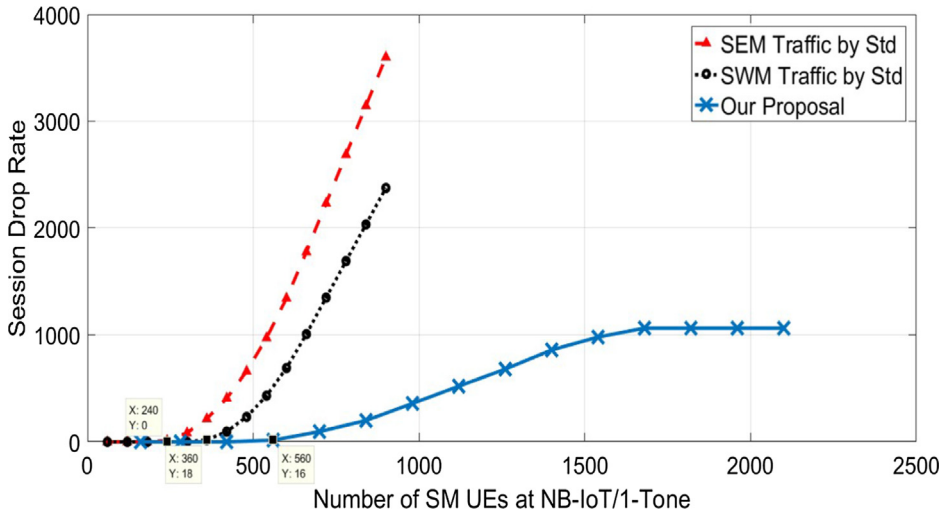


Fig. 18 Comparison of Session drop rate for current standard and the proposed scheme.

(2) The second main KPI is the number of served smart meter terminals from different utilities which simultaneously request access. We examined our optimization on one of the major applications, as the smart meters for two different utilities that serve the described geographical area. The number of SMs doubled after our proposal. Where, the served SMs increased from 720 terminals to 1680 terminals after the proposal. Only 20% of the eNB total bandwidth served this number of SM terminals. Also, this optimization was validated by RRC Setup Success percentage KPI. Where, we compared our optimization values with the model in [28]. Where, the lowest probability of channel busy was 9.37% and could serve only 5 IoT terminals per one channel with payload size 512Byte. However, our proposal and with the same calculation methodology served 7 SM terminals per one channel per second.

(3) The third main KPI is SDR. We achieved maximum served terminals with zero session drops reached to 560 SMs of 1560 SMs of RRC setup success. Our proposal enhanced the session drop by 35.7% of only 20% of the eNB total bandwidth, referring to the session drop probabilities in reference [29]. Where, in reference [29], the session drop probability was 23% for 60% NB-IoT static reservation from total eNB bandwidth. So, their session probability will increase by 69% if they used only 20% of eNB total bandwidth for NB-IoT. However, our proposal achieved only 33.33% session drop probability with only 20% of eNB total bandwidth for NB-IoT.

Whatever the payload increase or uses of other techniques, many radio frames are still not effectively utilized (even when they will be allocated later). A comparison of our records with the previously mentioned methods shows the ability of our approach to utilize the resources wasted for signalling and higher layer control messaging at each transmission process and to maximize the utilization of radio frame resources. Therefore, our proposed approach achieves the following:

(1) On the IoT terminal side, it reduces the signalling and access request burden on SM terminals at each transmission request,
(2) On the NB-IoT eNB side, it enhances the system spectral efficiency,
(3) At both the smart meter terminal and NB-IoT eNB, it will result in a positive impact on energy efficiency,

Well scheduling for radio frames, avoid the accumulative delay due to multiple trials of random access requests until success to acquire data channel even with using alternative techniques as D2D [37].

4. Conclusion

The future increasing demands' for smart meters businesses require maximizing the resources utilization to serve maximum possible numbers of NB-IoT UEs. Additionally, the traffic management subject over NB-IoT for efficient spectrum is a newly opened issue at IoT 5G. In this paper, we achieve two primary goals. The first one is to improve the NB-IoT spectral efficiency to serve the increasing numbers of smart meters. The second one is to reduce the signalling burden during each transmission request for smart meters businesses. Therefore, this paper introduces a new integrated approach to enhance the NB-IoT spectral efficiency that serves the smart meters applications. First, we model the protocol stack of the NB-IoT access network by using the state-machine modelling methodology on the Simulink environment. According to our goals, we focus on the NB-IoT uplink scheduler in MAC layer model. For fair results, we divide the cell spectrum of NB-IoT equally on the four data rates. Also, we assume ideal propagation without any transmission impairments. The simulation results are verified according to 3GPP standard and are validated with other trusted of different modelling technique. Second, we optimize the NB-IoT uplink scheduler, which exploited the periodicity nature of smart meters. We executed that by proposing an integrated scheduling protocol which rearranges the transmission times

of different smart meters utilities and draws a map for the transmission schedule of them. Accordingly, the comprehensive utilization of free time resources achieves. We applied the optimization on a case study for a designated geographical location has smart meters for two different utilities. The proposed approach encompasses three visions that comprise signalling, architecture, and algorithm. Each vision represents an update on in current NB-IoT 3GPP standard. The optimization is evaluated in term of the number of smart meters terminals versus four KPIs; resources utilization, spectral efficiency, RRC setup success percentage, and session drop rate. Finally, the simulation results prove the ability of our proposal to increase the spectrum utilization to 17.47% according to the used traffic profiles. This enhancement reflects on spectral efficiency that improved two times.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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